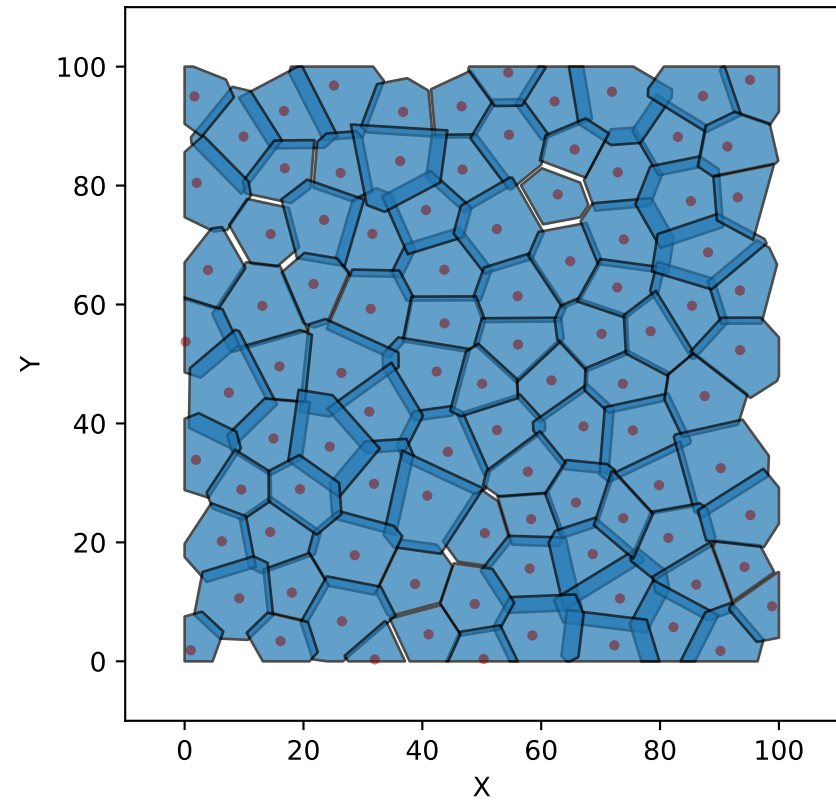
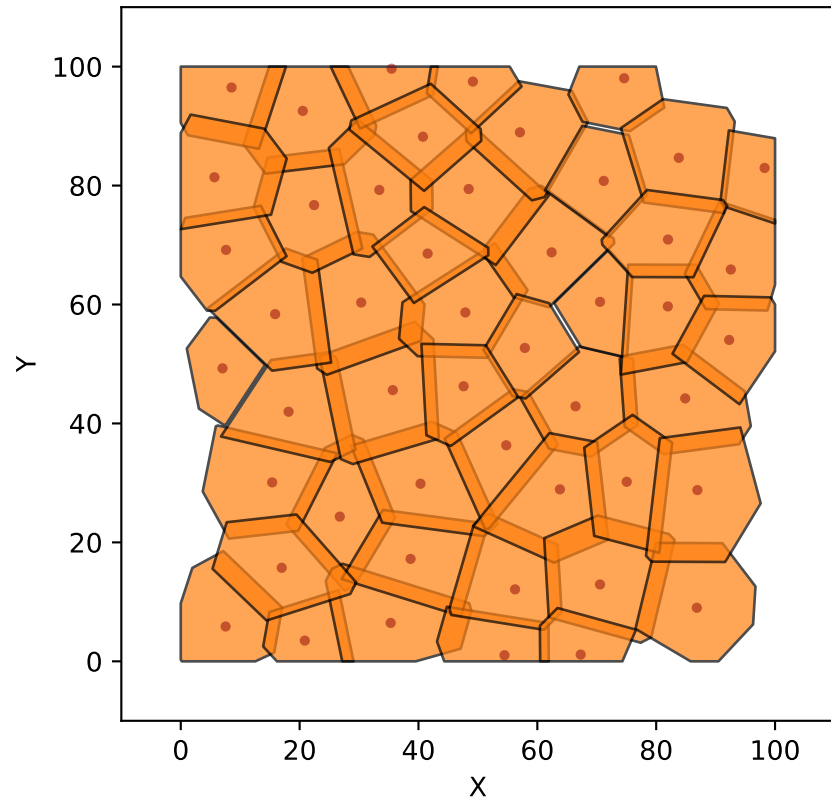


Cluster 0
99 cells

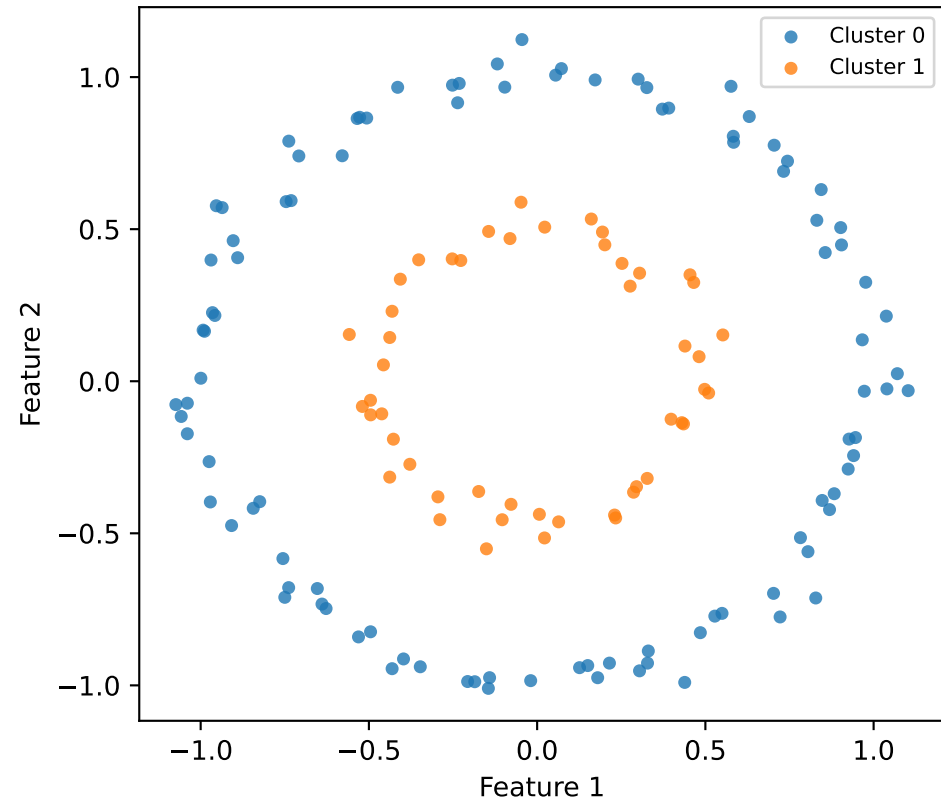


circles — ground truth

Cluster 1
49 cells

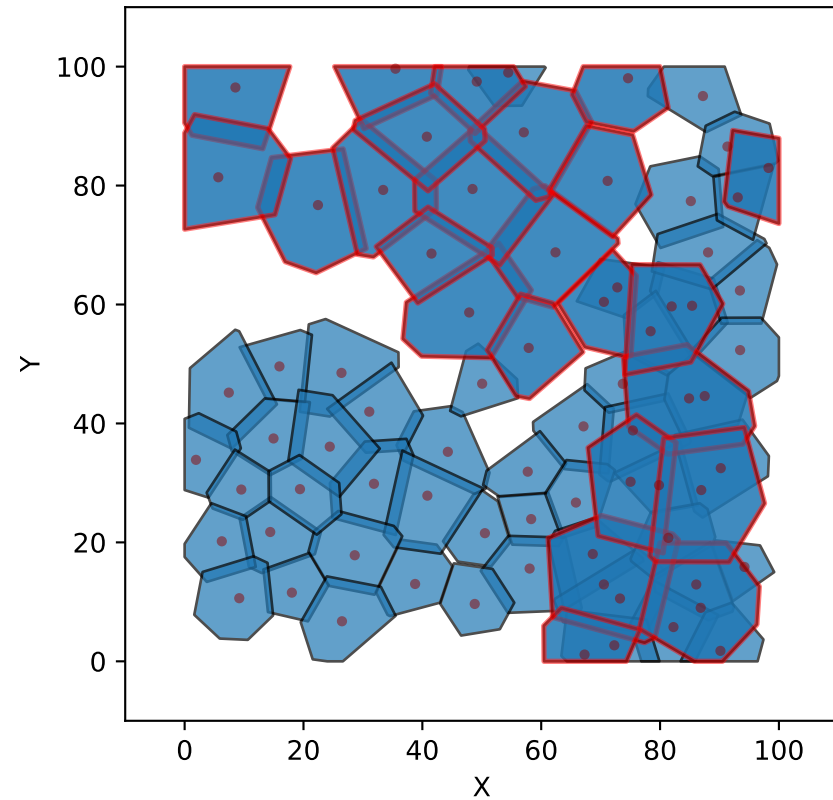


Feature space

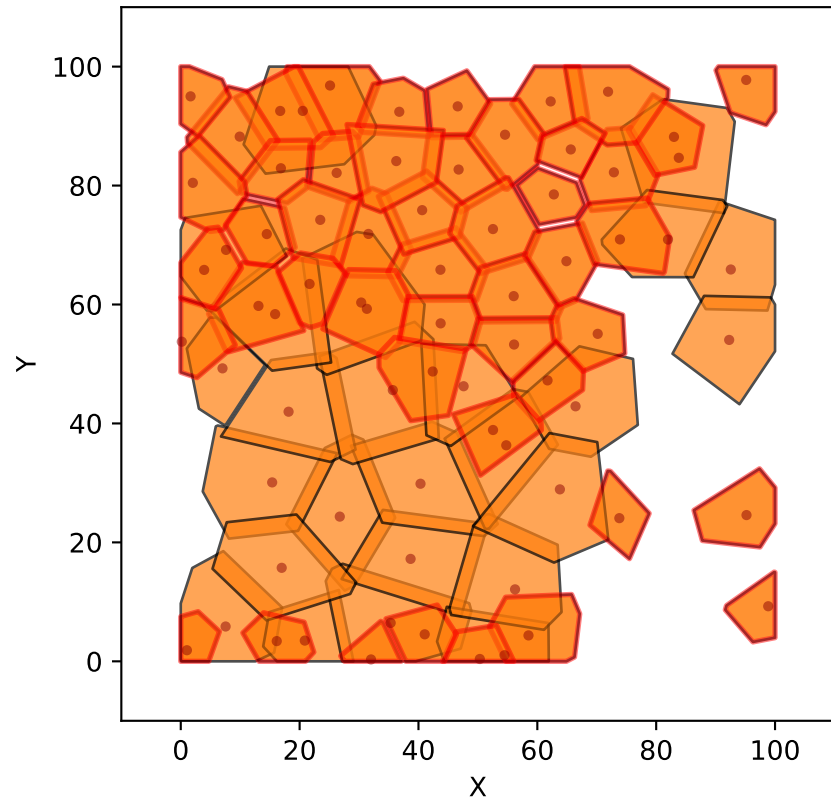


circles — GMM + MRF ARI=-0.005 violations 142 -> 106 changed=17 iters=4

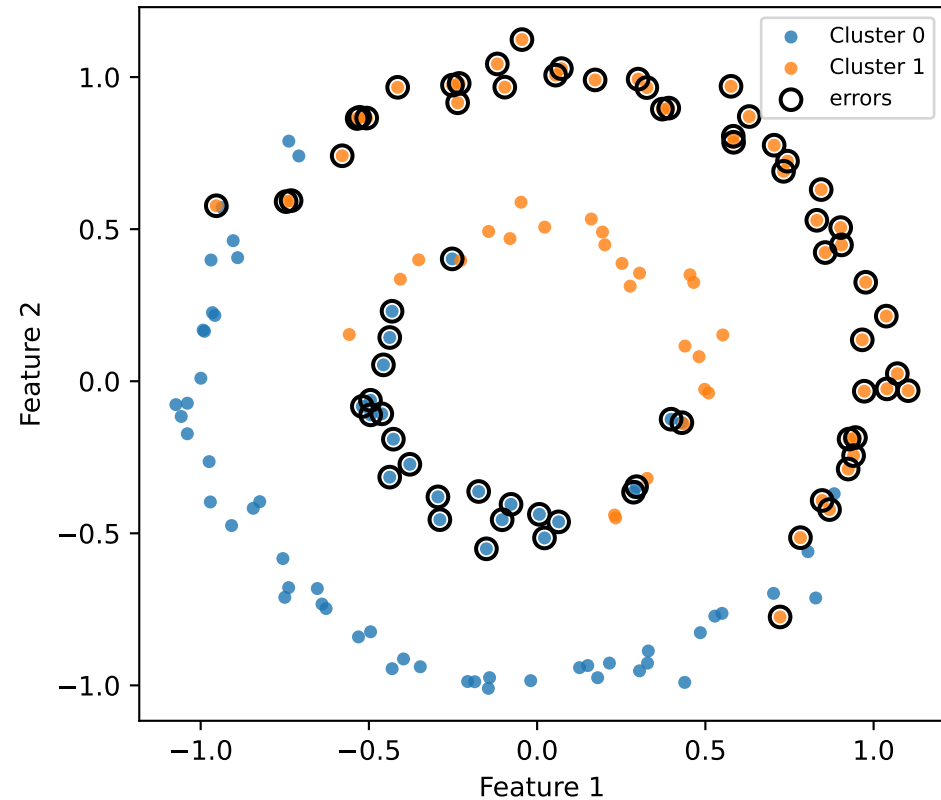
Cluster 0
75 cells (24 err)



Cluster 1
73 cells (48 err)

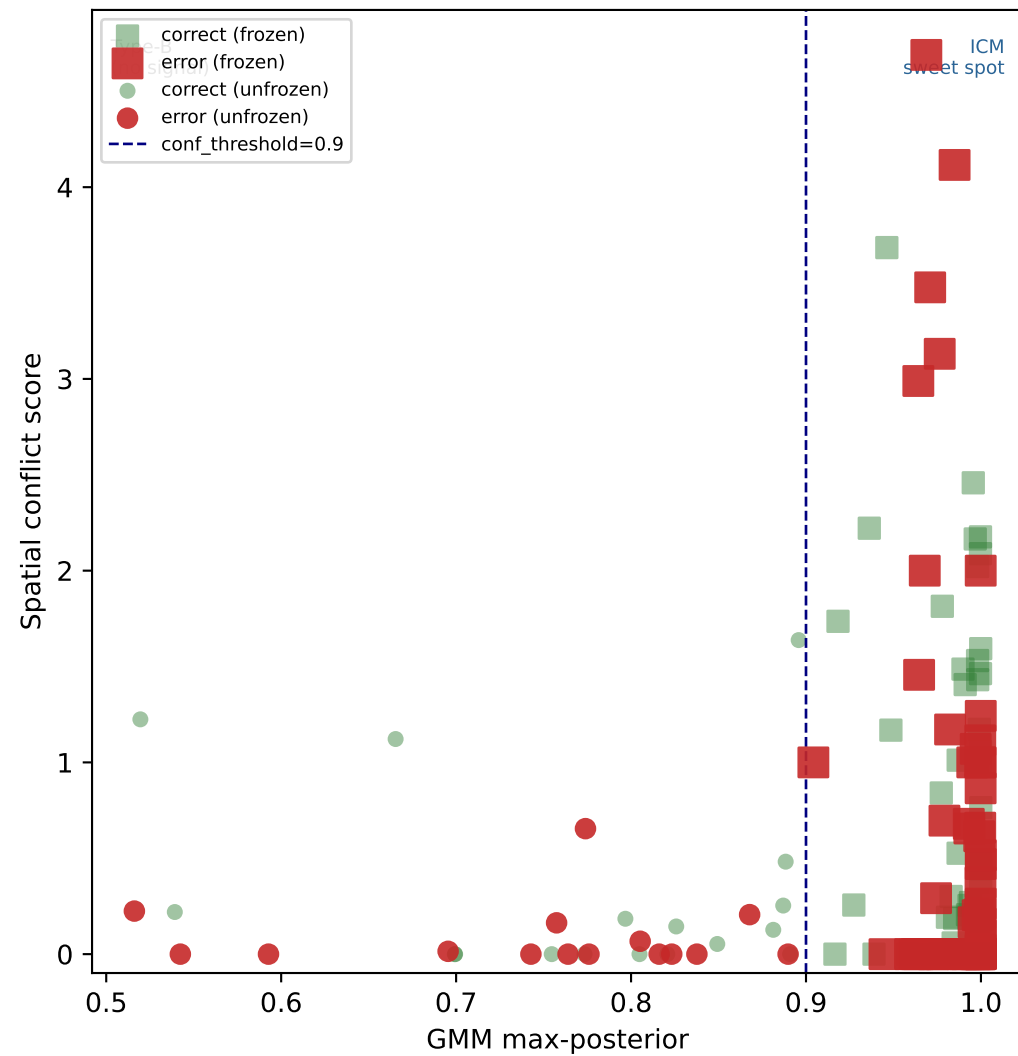


Feature space

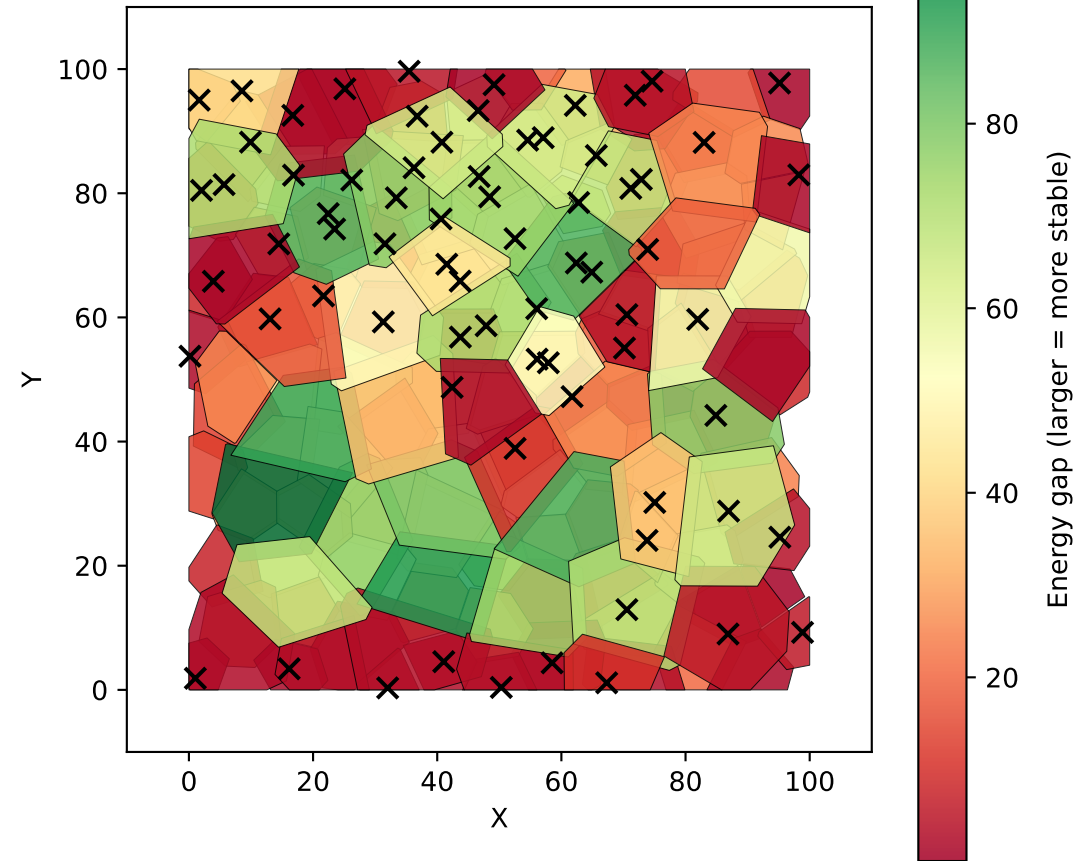


circles — MRF diagnostics

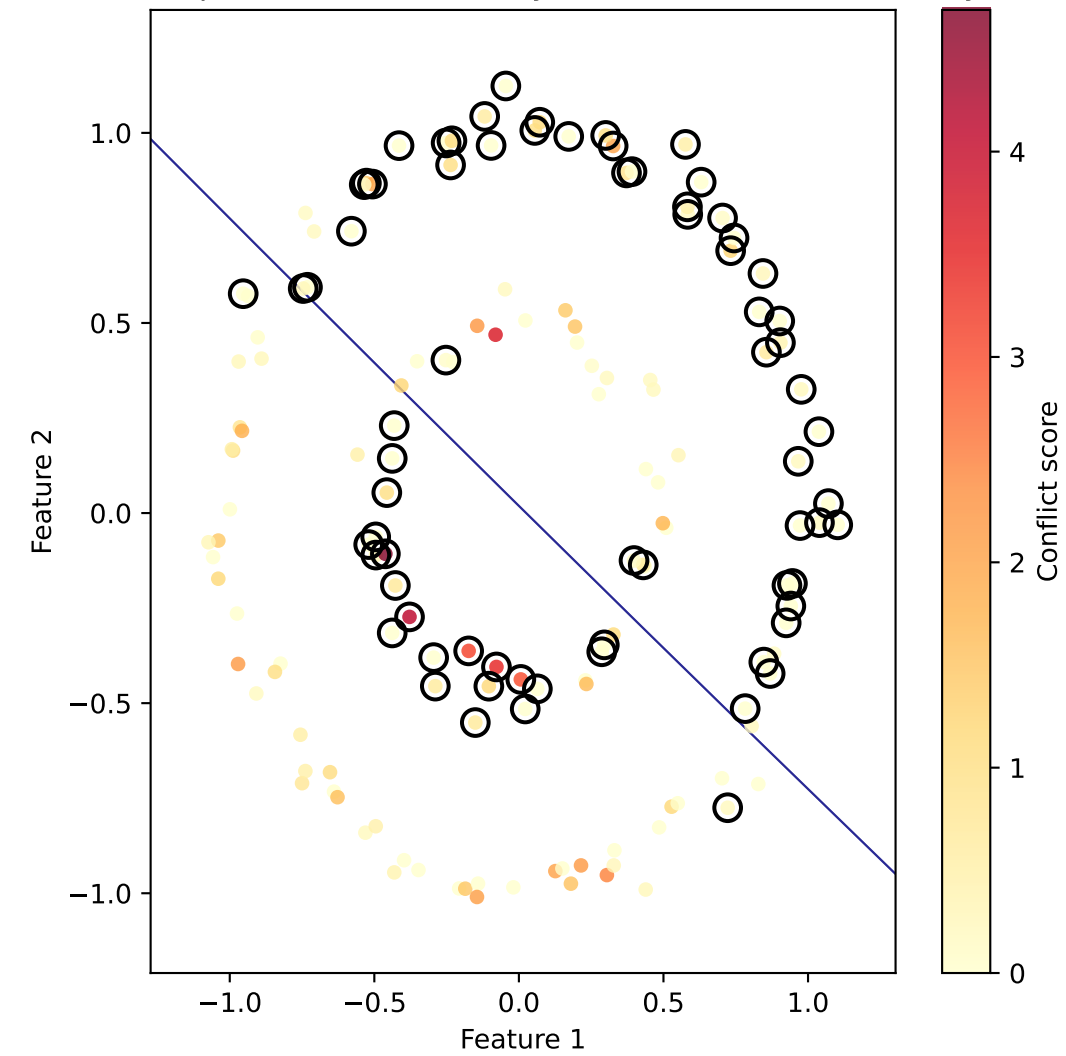
Conflict-confidence scatter



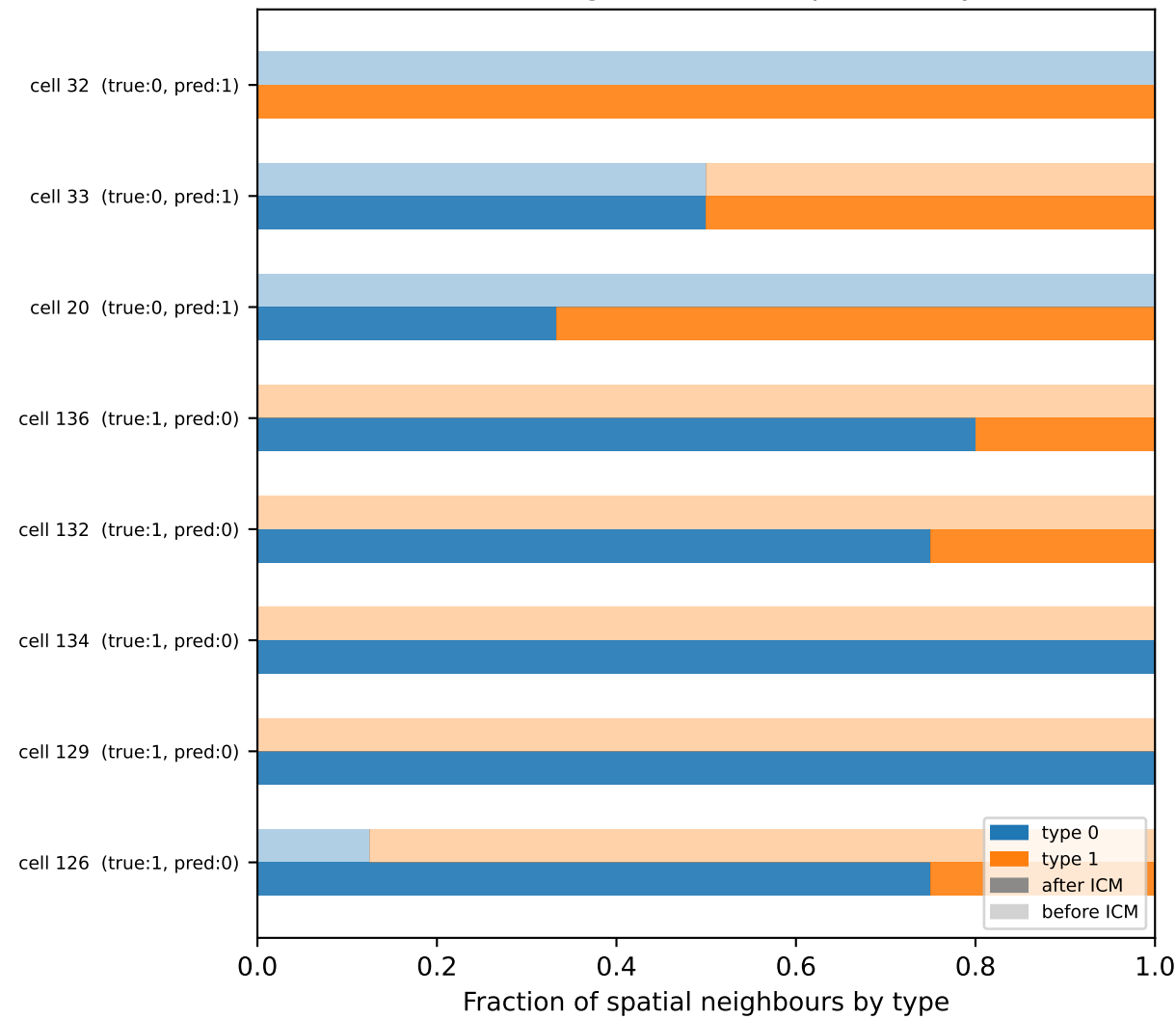
Spatial energy-gap map (x = remaining error)



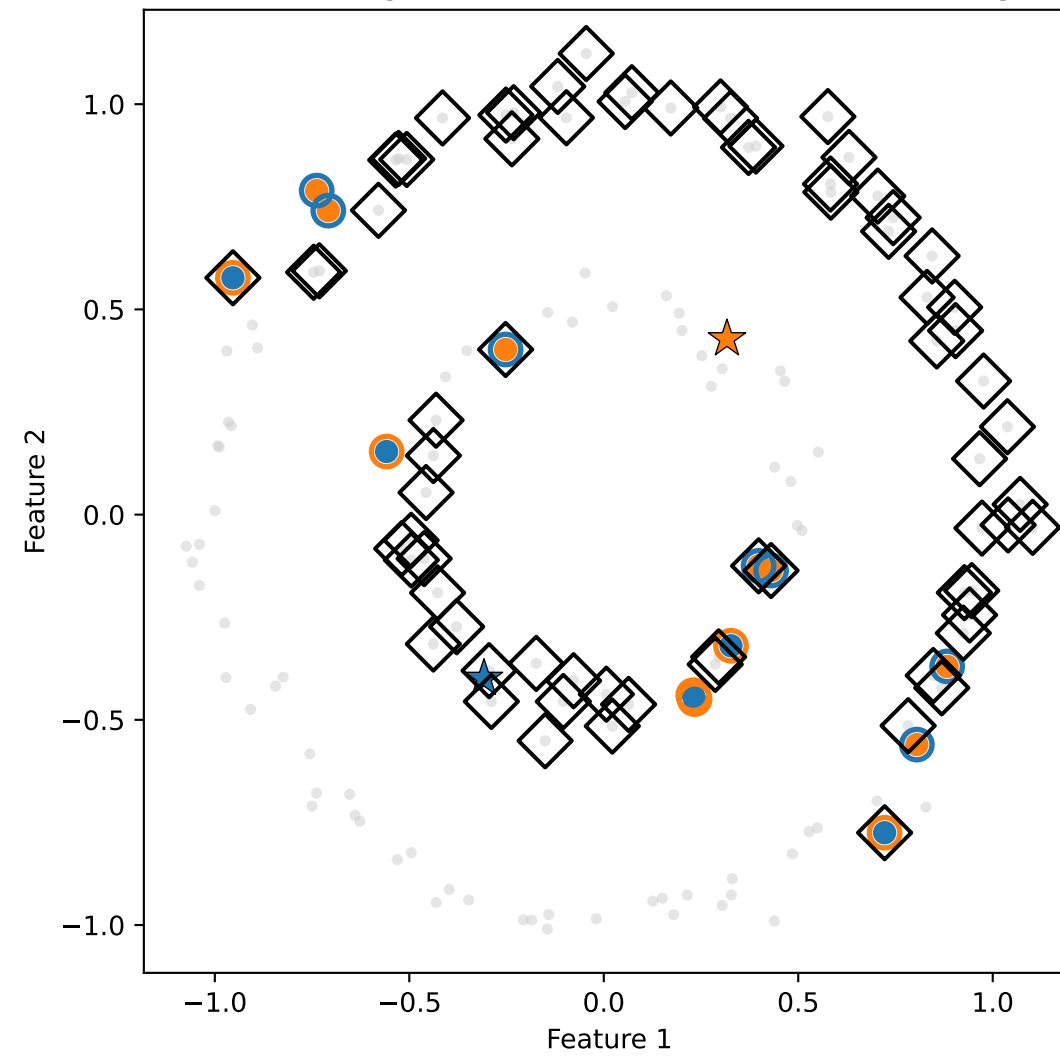
Feature space + conflict (navy = GMM decision boundary)



Neighbour type composition (dark=after ICM, light=before; top errors by conflict)



ICM label changes: 13 cells moved (fill=old label, ring=new; ★=cluster mean; ◇=remaining error)

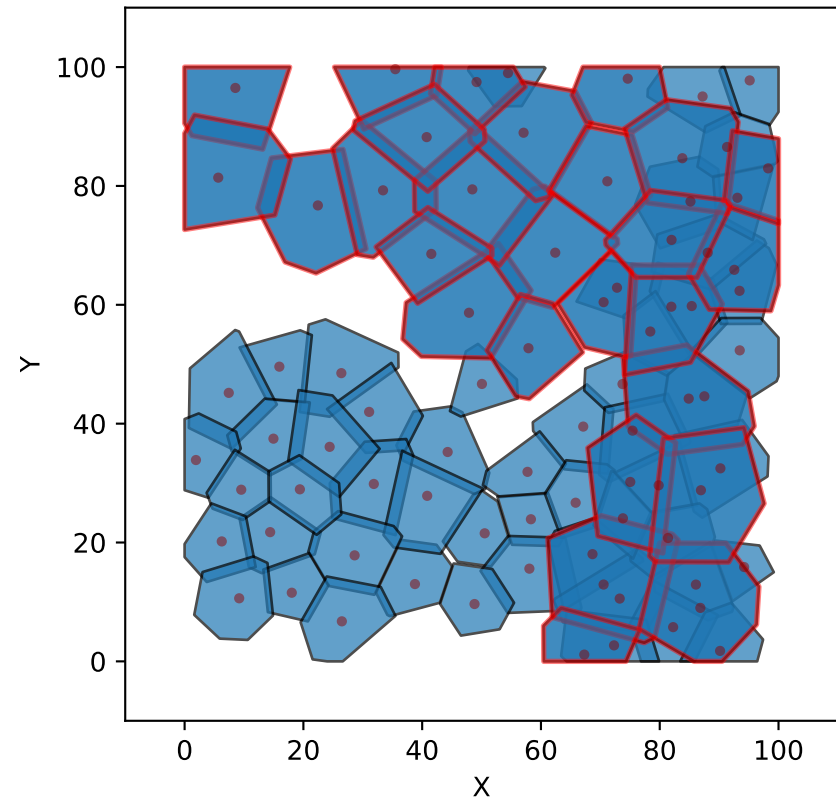


Run summary

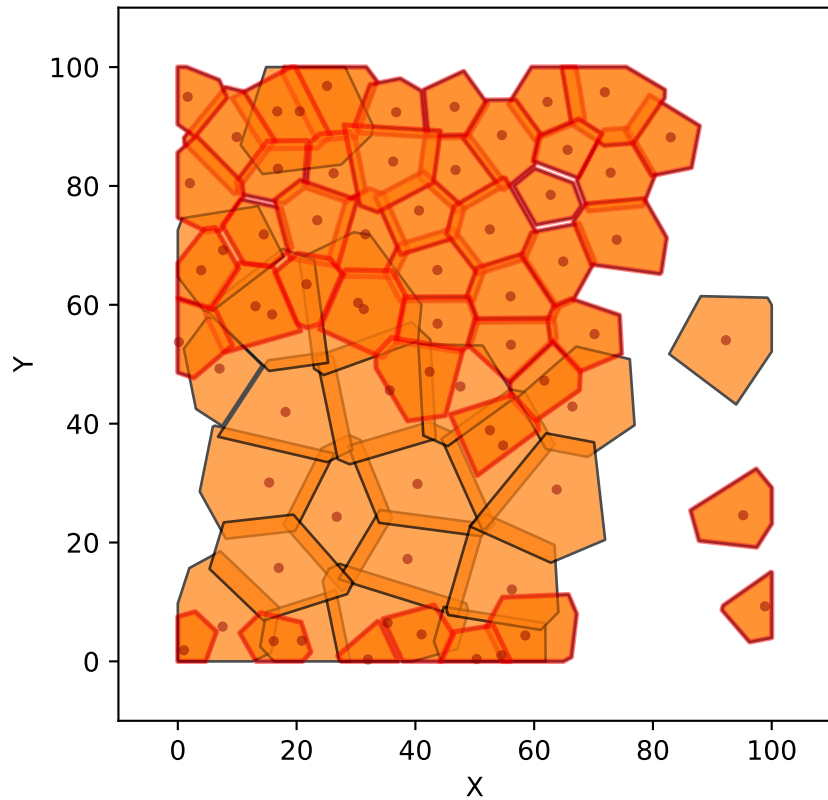
MRF Run Summary	
ARI:	-0.0053
errors:	72 / 148
violations:	142 → 106
n_changed:	17
n_iters:	4
n_frozen:	116
Parameters:	
lam:	20.0
tau_low:	0.1
tau_high:	0.4
radius:	15.038613891552956
conf_thr:	0.9

circles — GMM + MRF (signed ramp, tau_low=0.3) ARI=-0.007 violations 60 -> 35 changed=10 iters=2

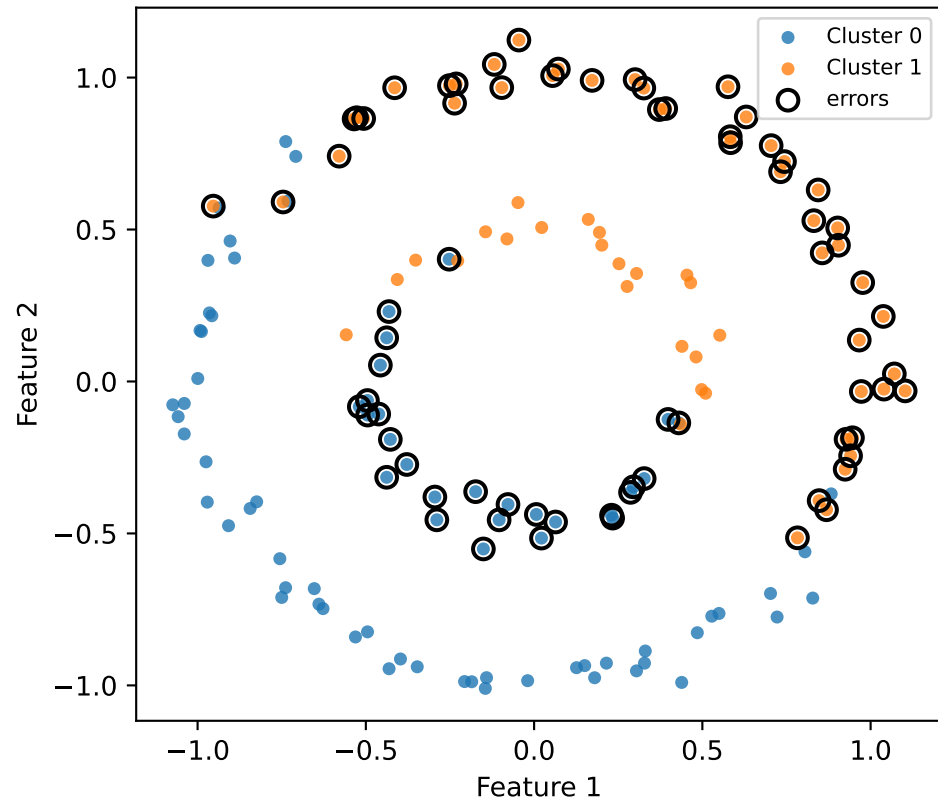
Cluster 0
80 cells (27 err)

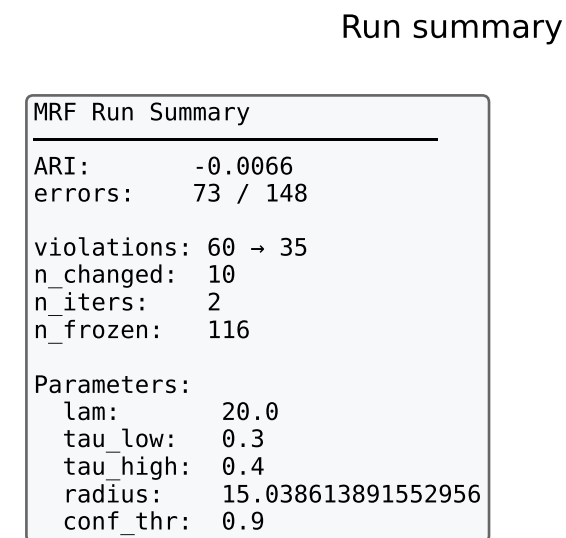
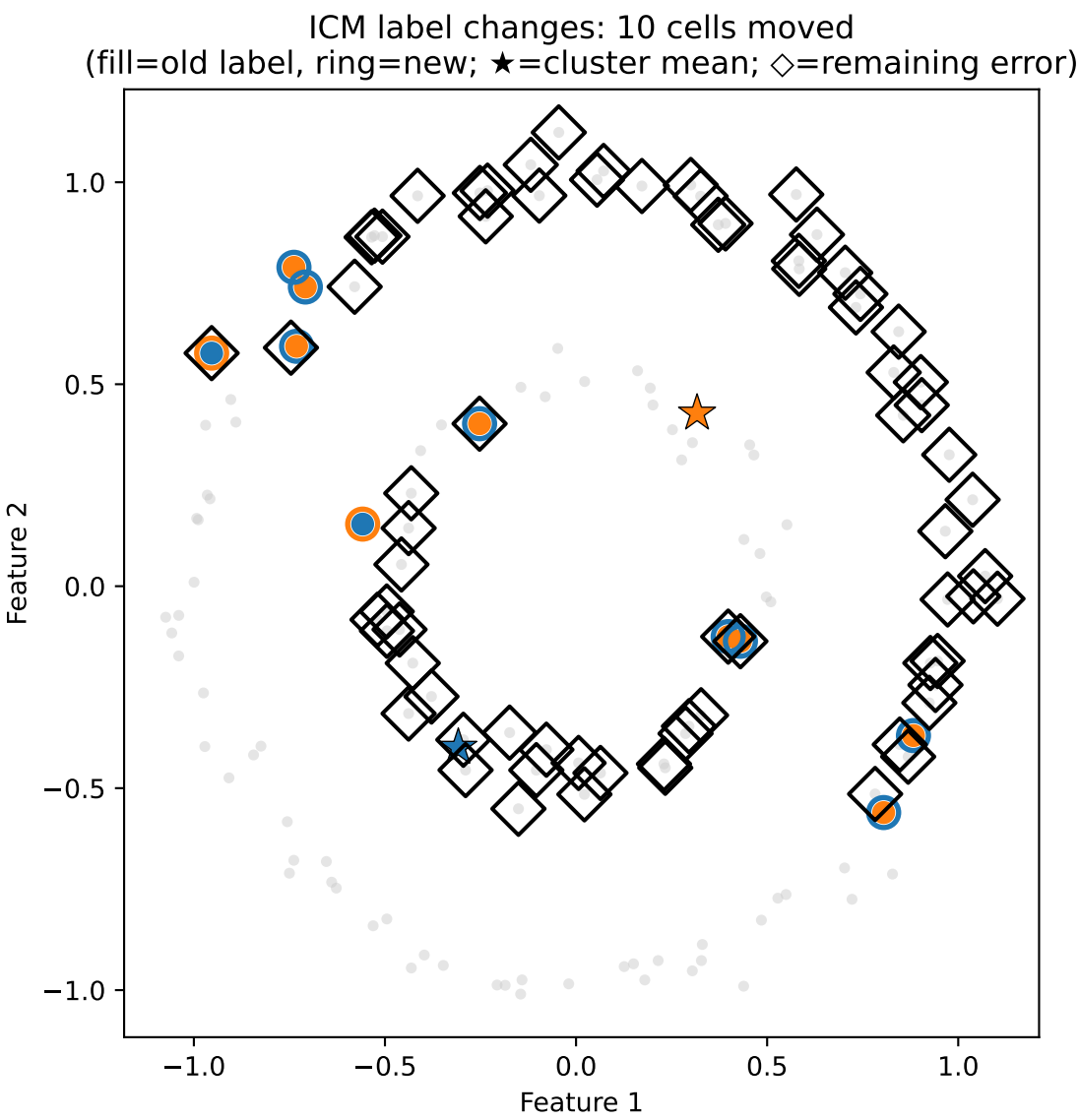
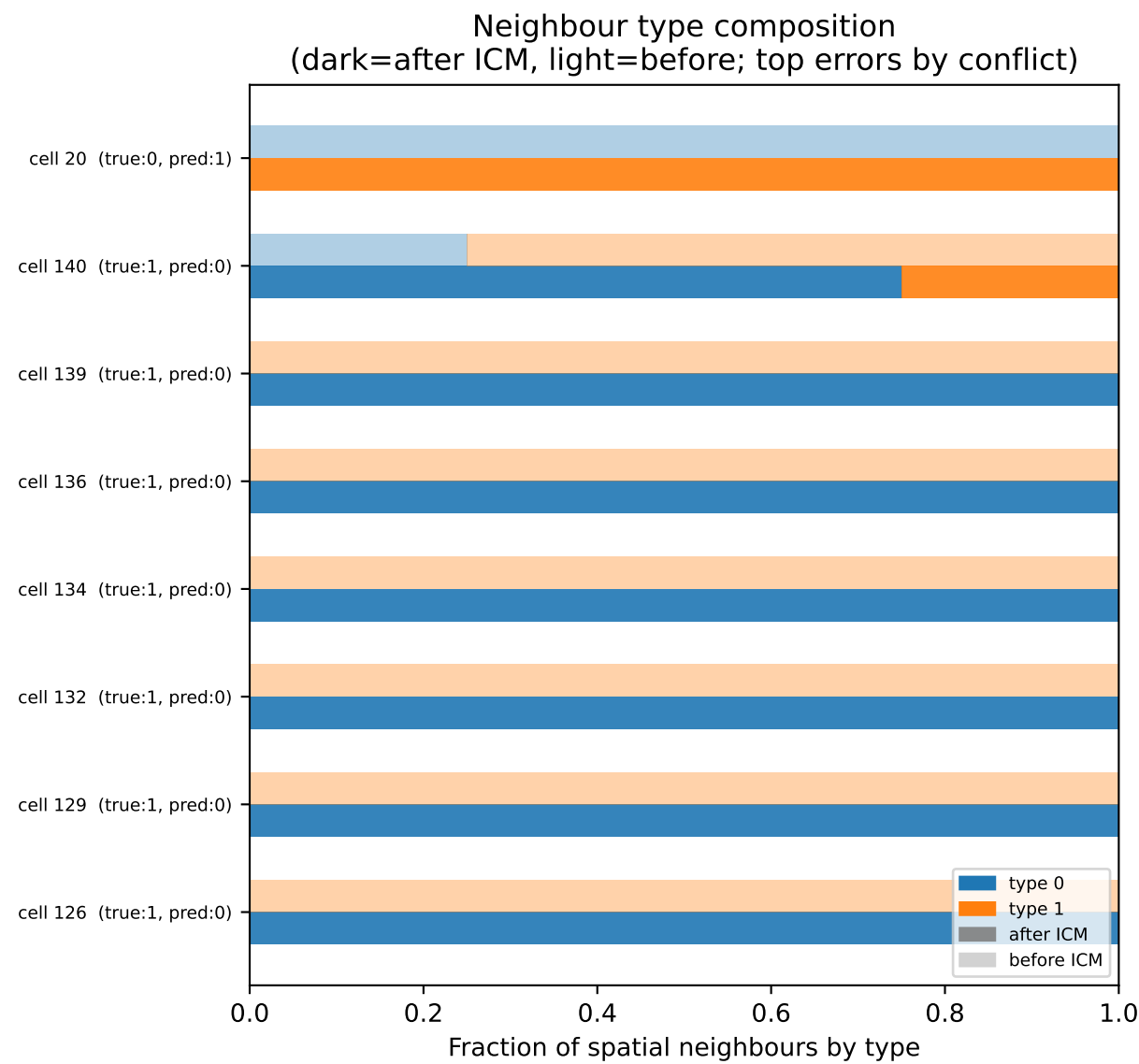
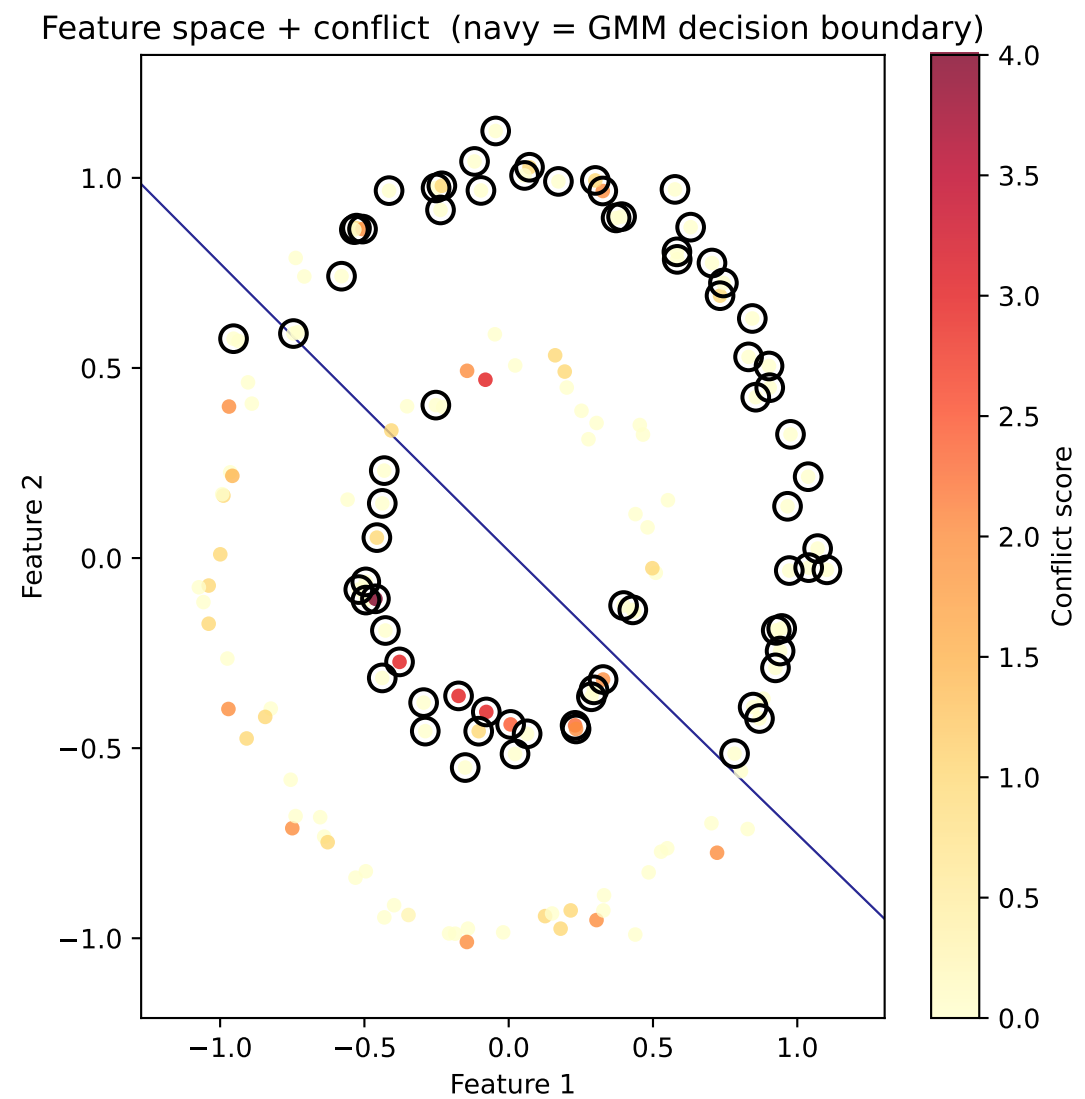
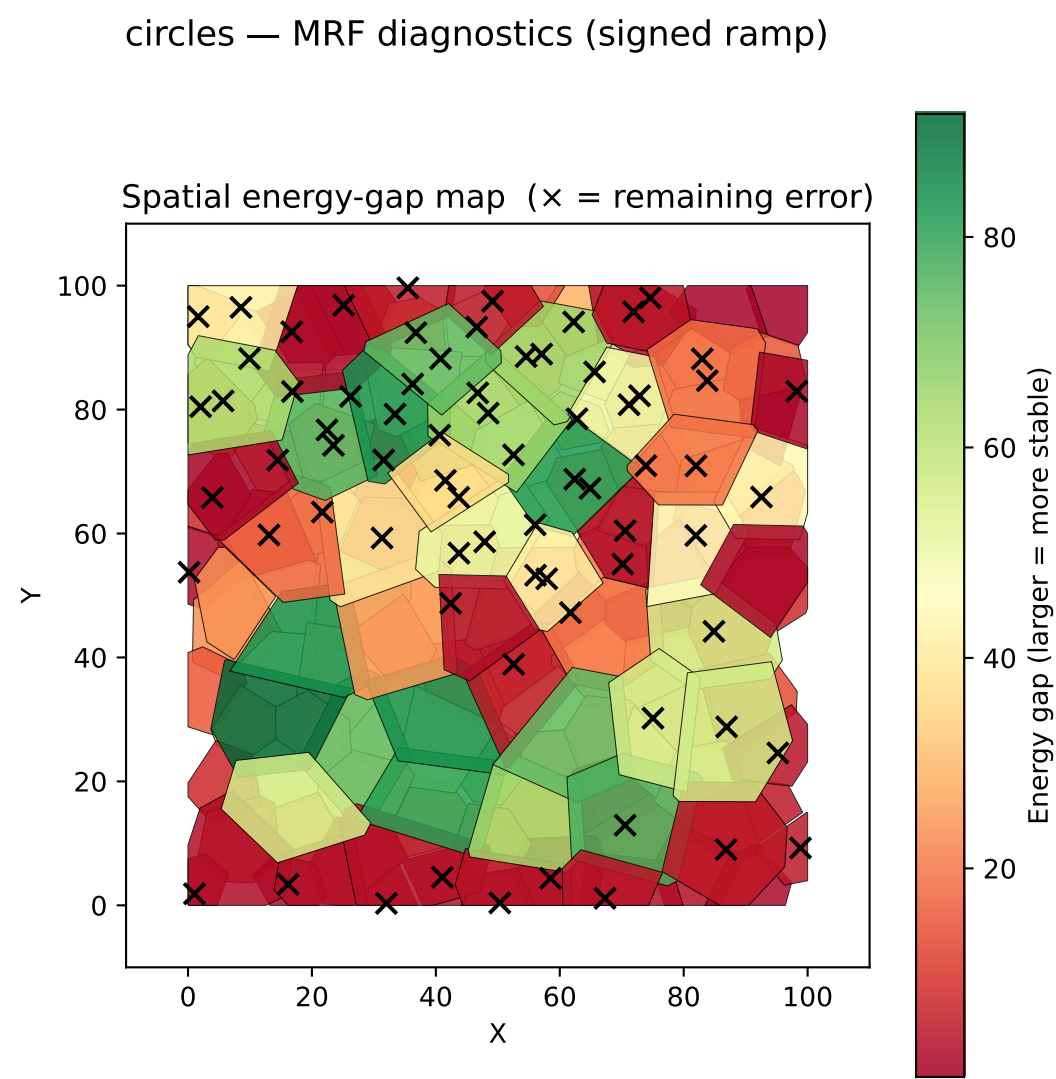
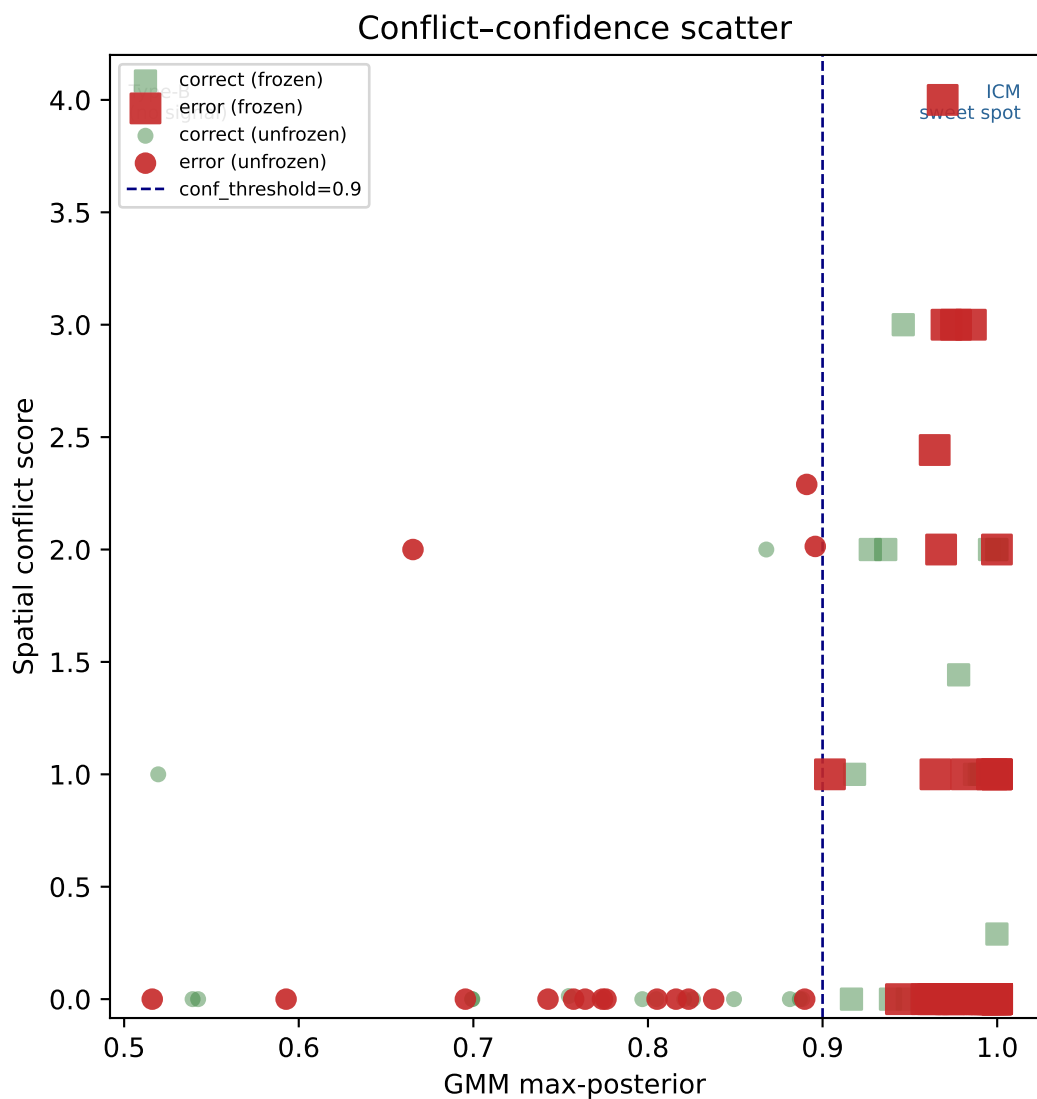


Cluster 1
68 cells (46 err)



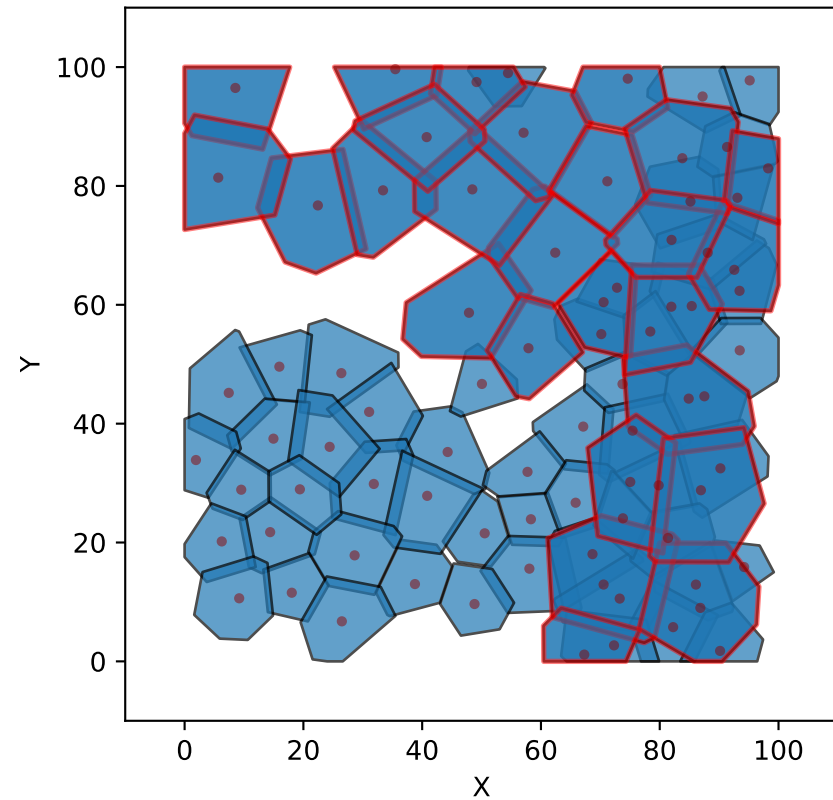
Feature space



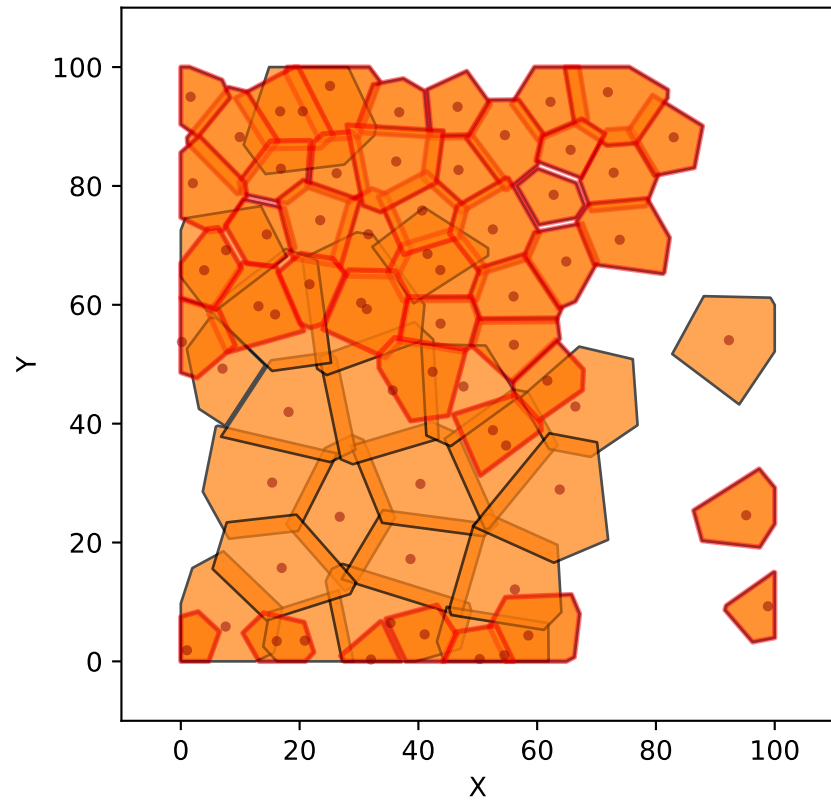


circles — GMM + MRF (signed log-ramp $\alpha=10$, $\tau_{\text{low}}=0.3$) ARI=-0.005 violations 60 -> 38 changed=12 iters=3

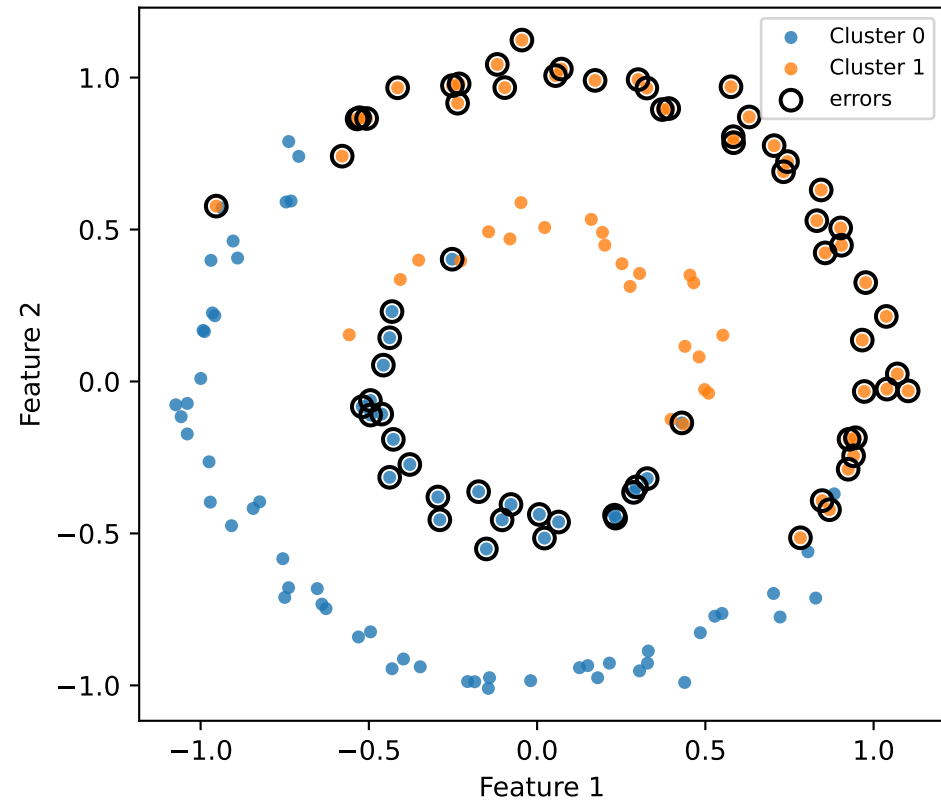
Cluster 0
80 cells (26 err)

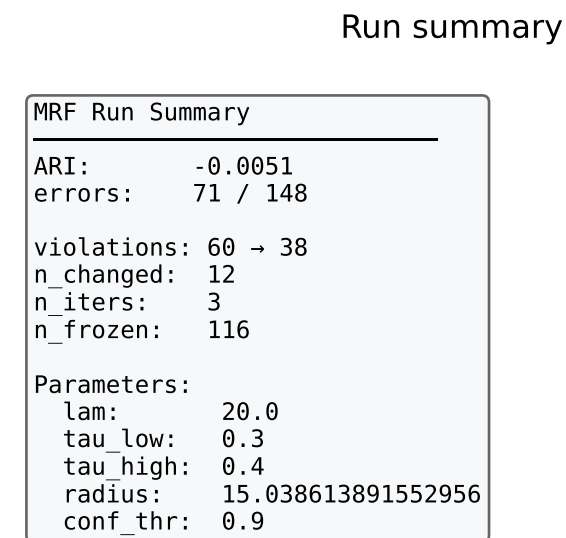
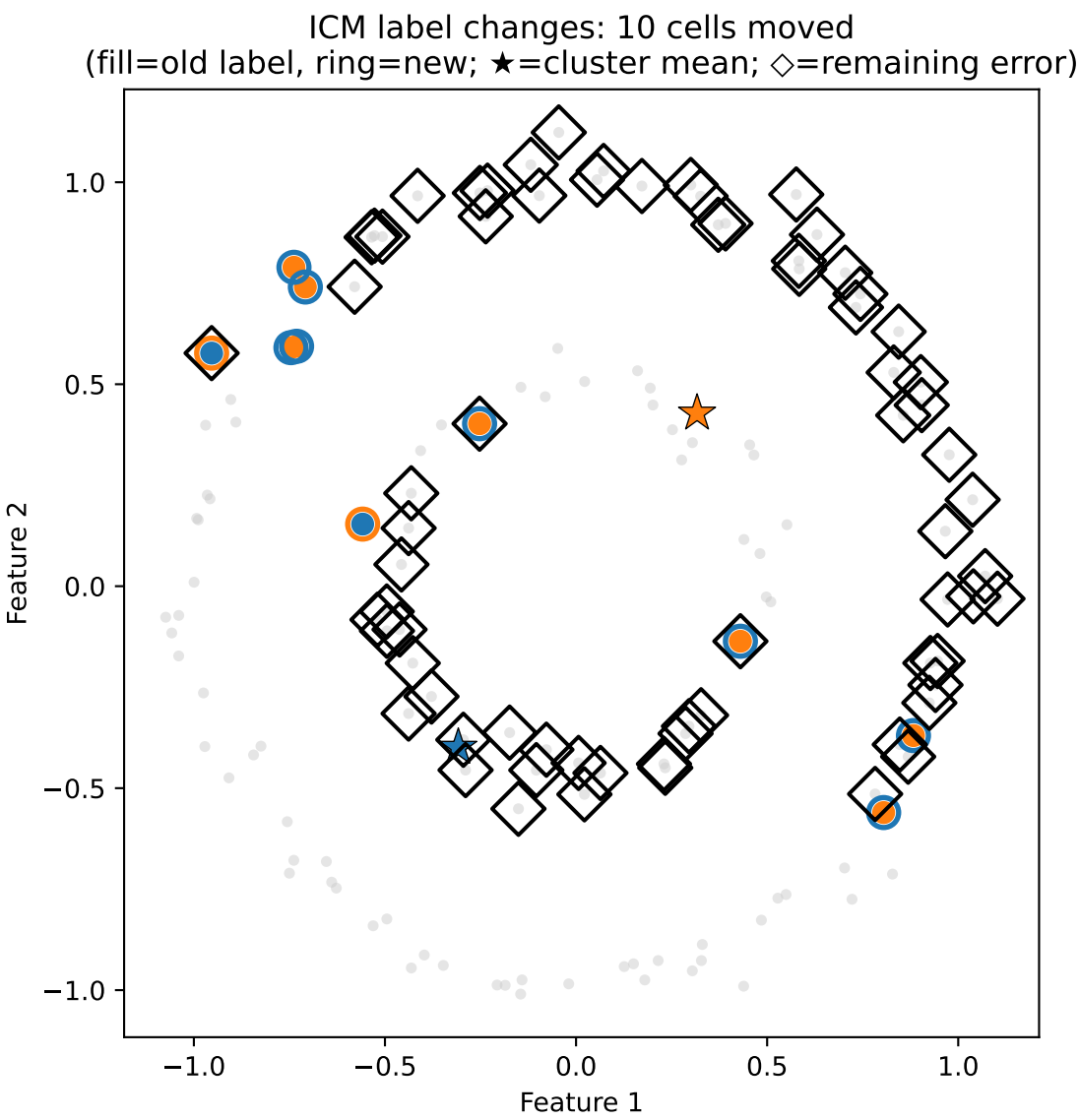
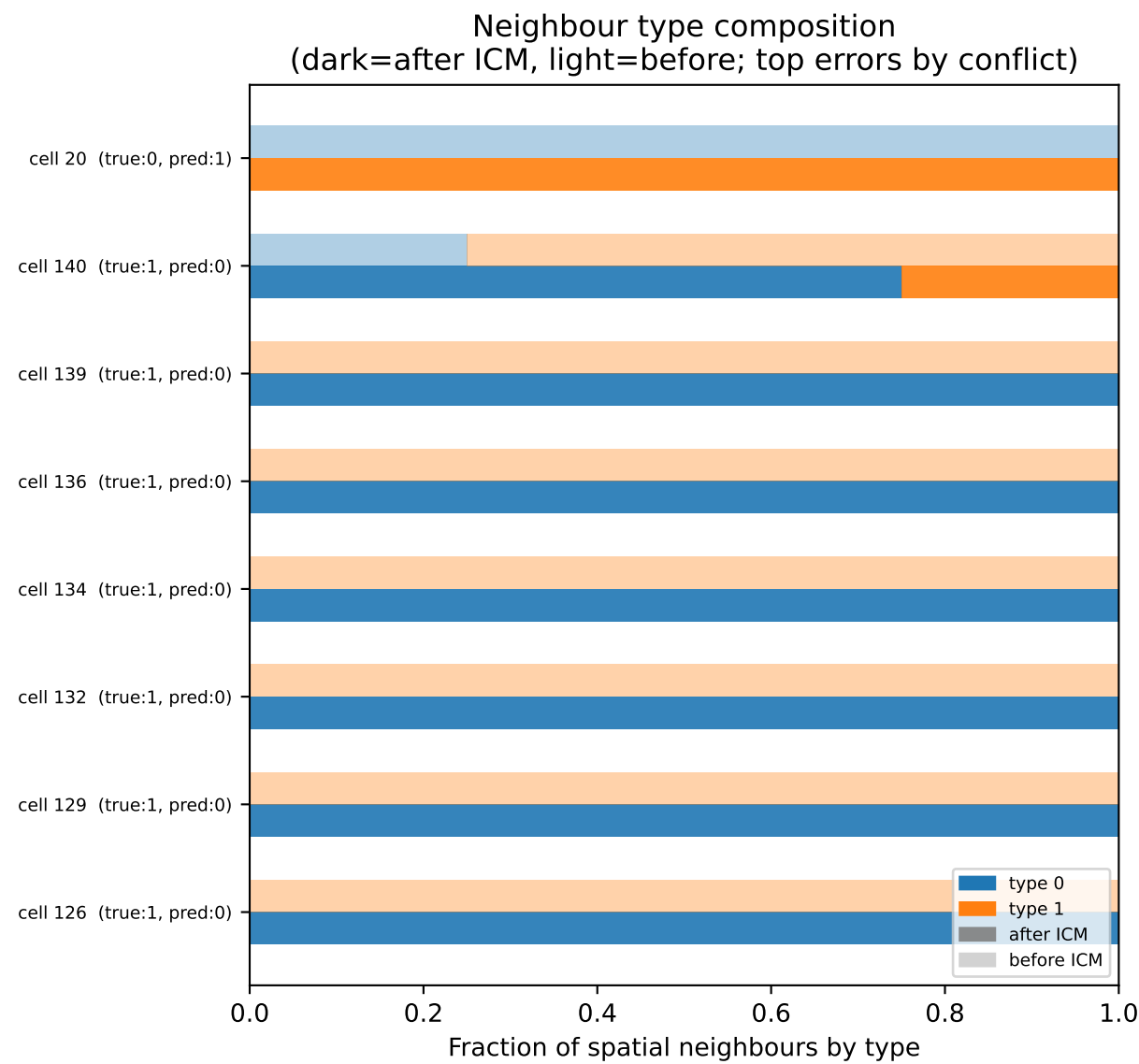
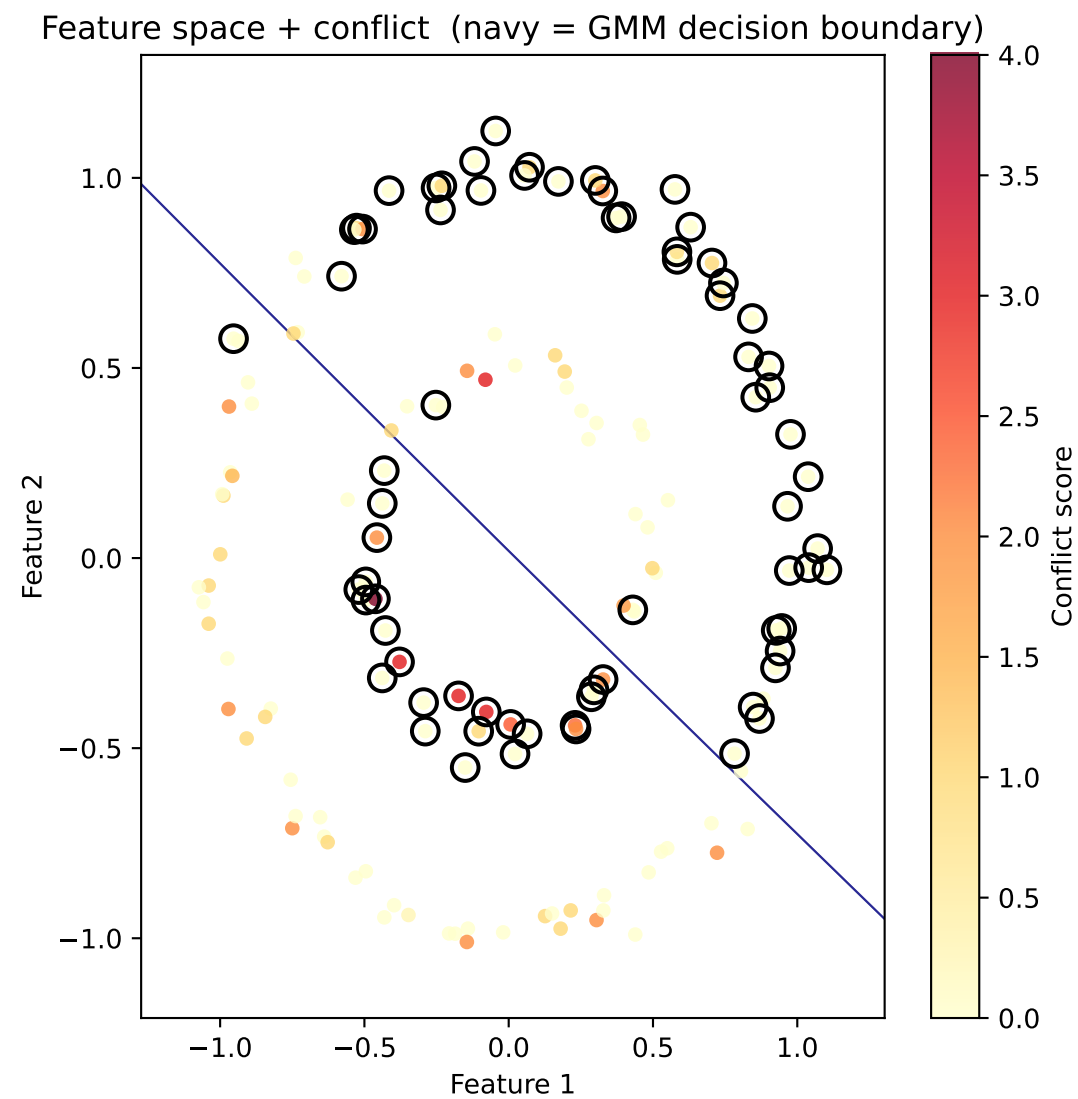
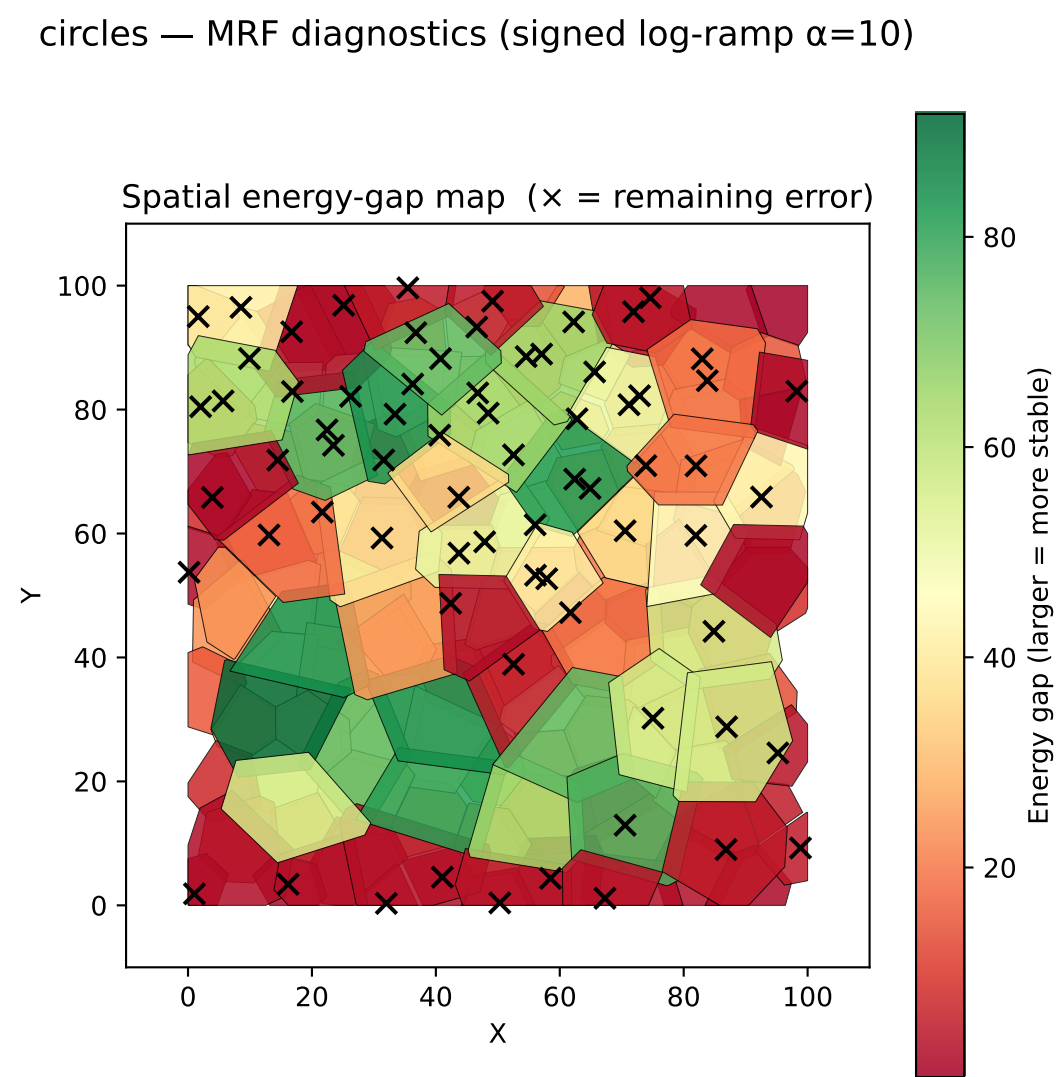
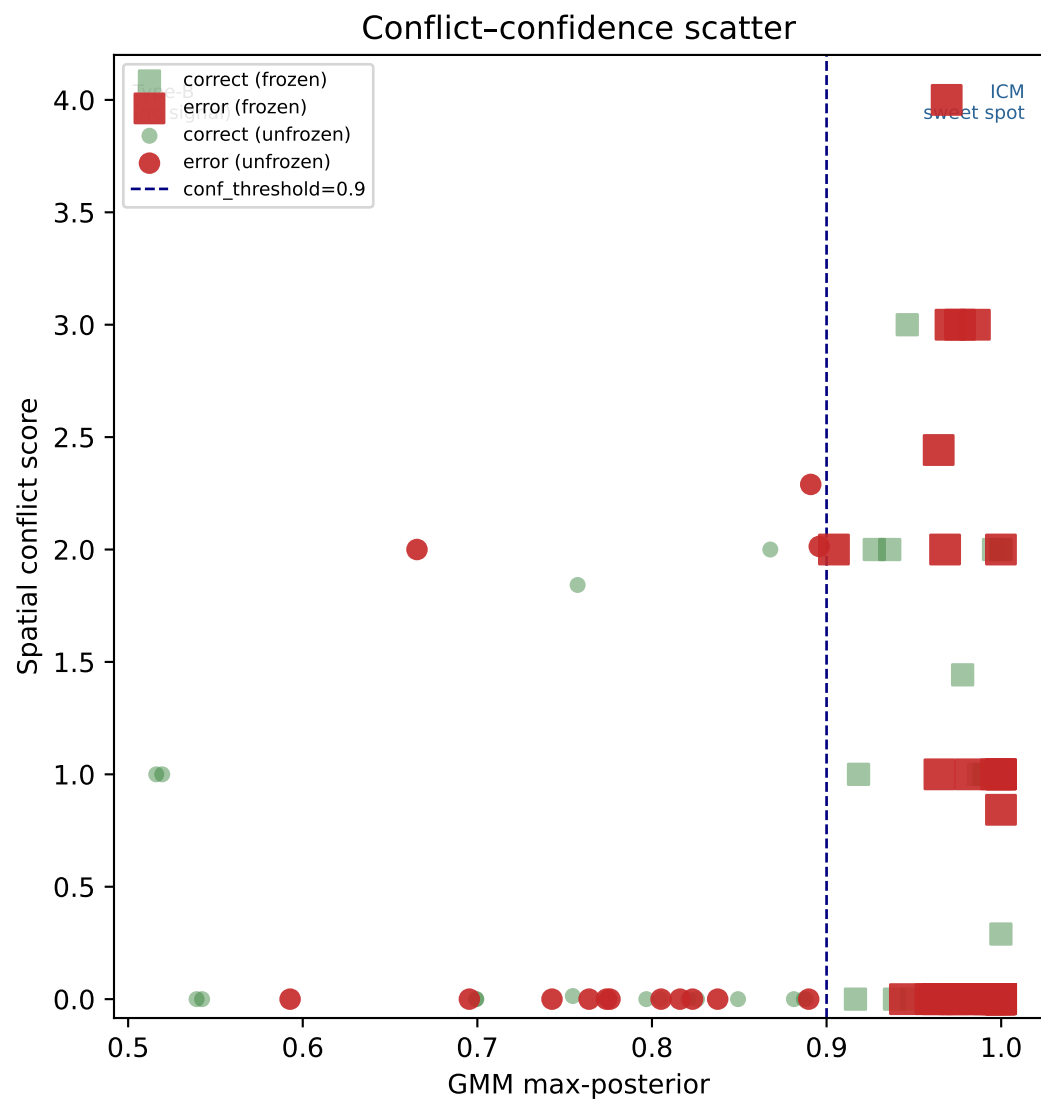


Cluster 1
68 cells (45 err)



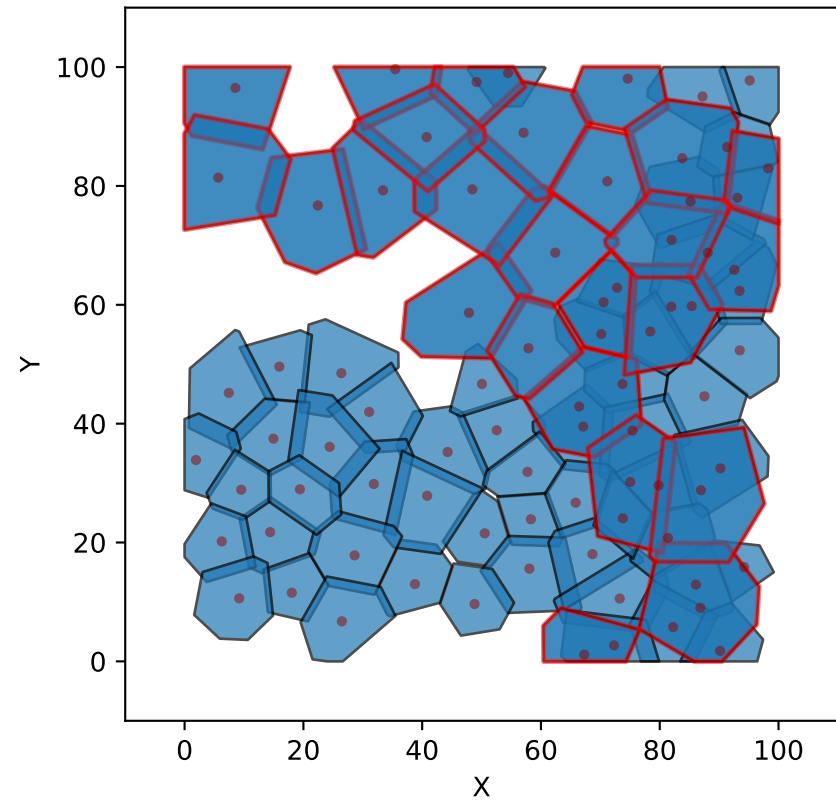
Feature space



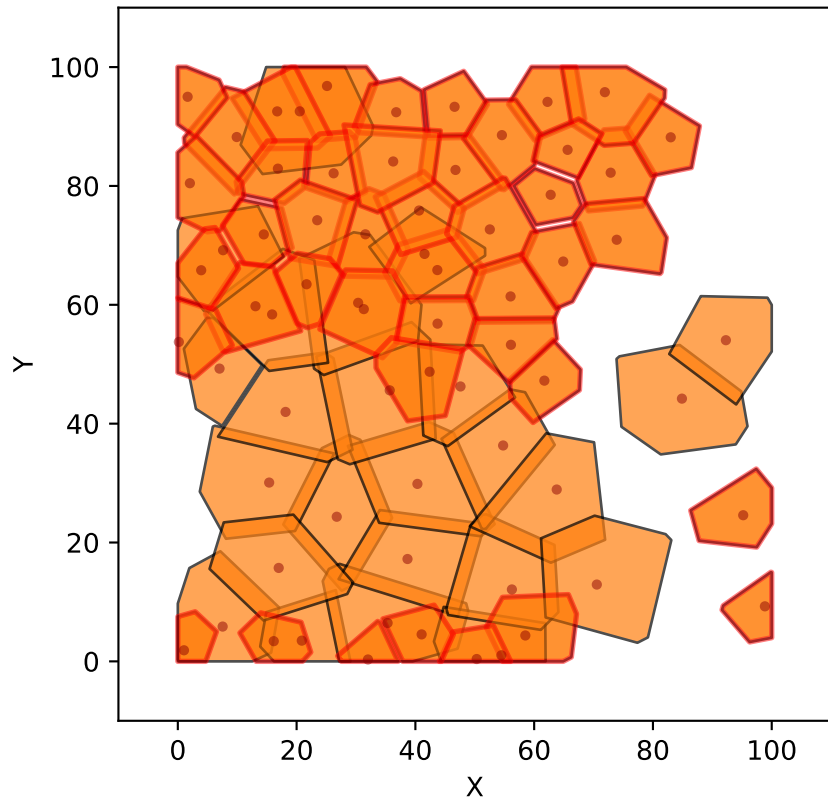


circles — GMM + MRF (signed log-ramp a=10 + H1 th=0.85) ARI=-0.002 violations 60 -> 31 changed=14 iters=2 force_unfrozen=5

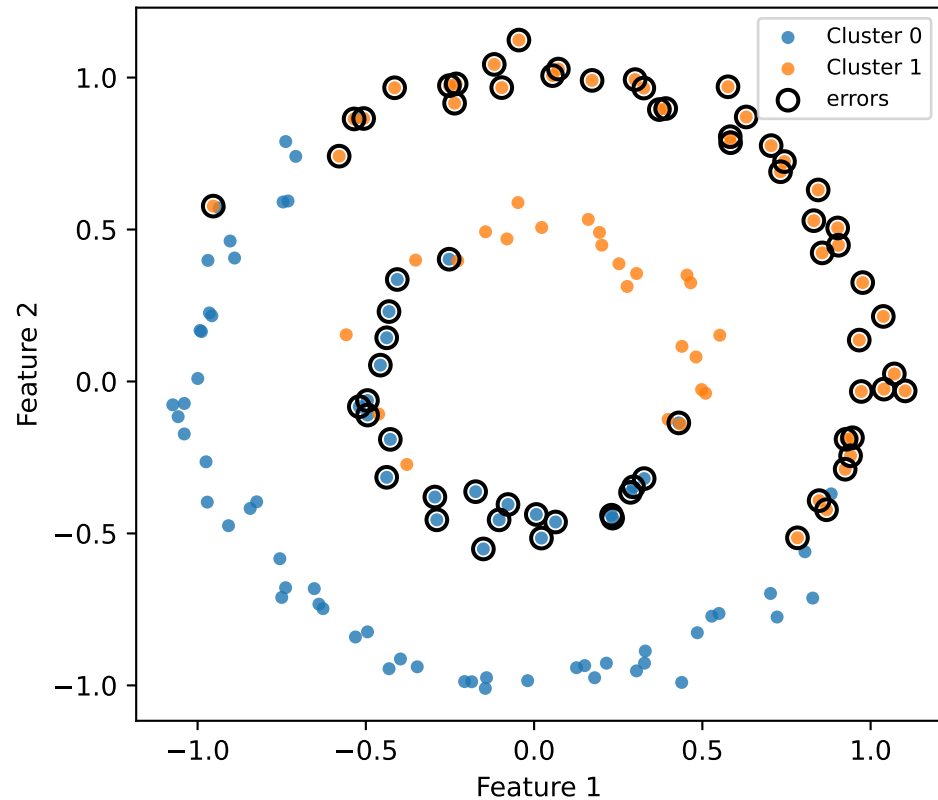
Cluster 0
80 cells (25 err)

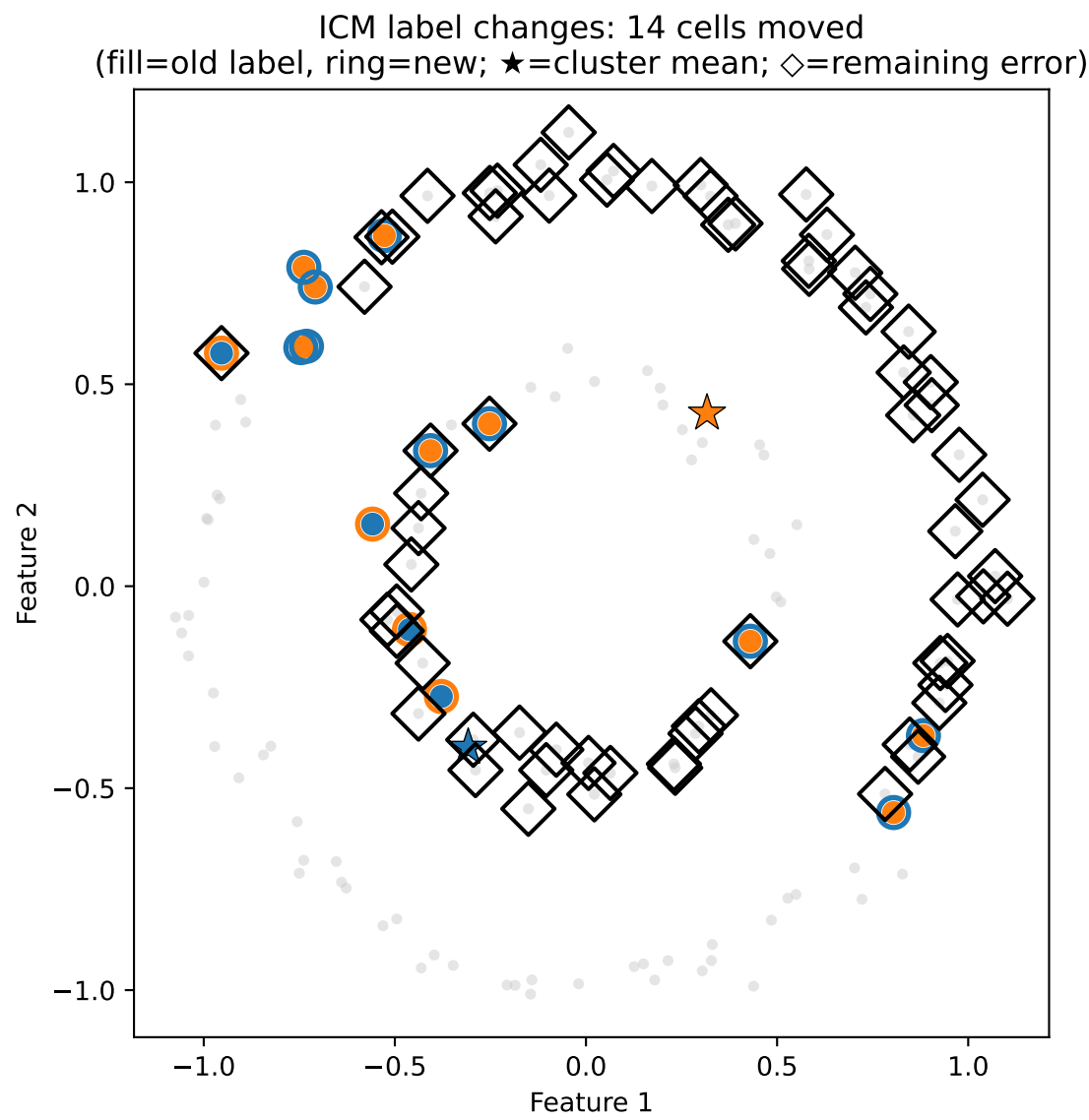
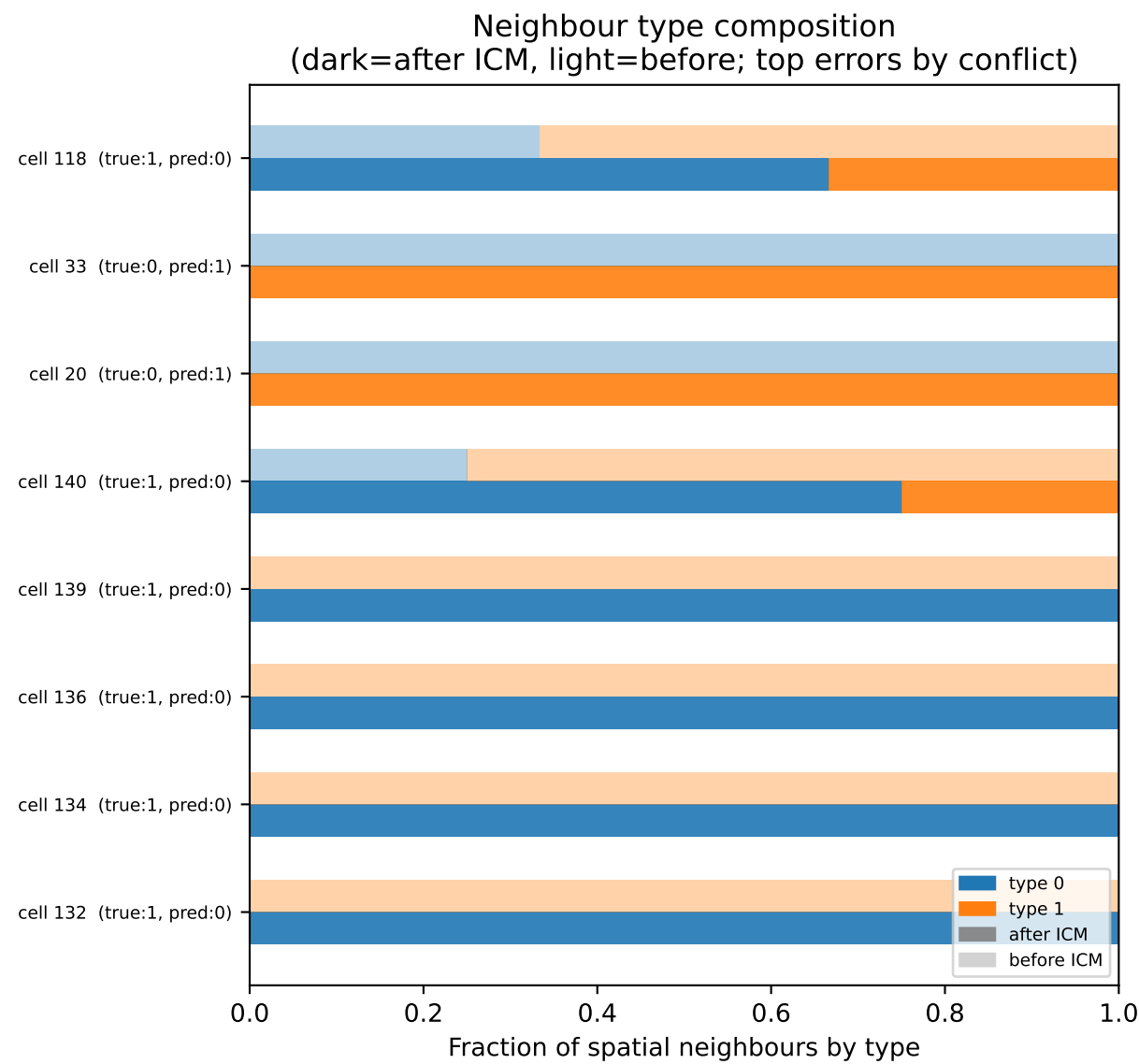
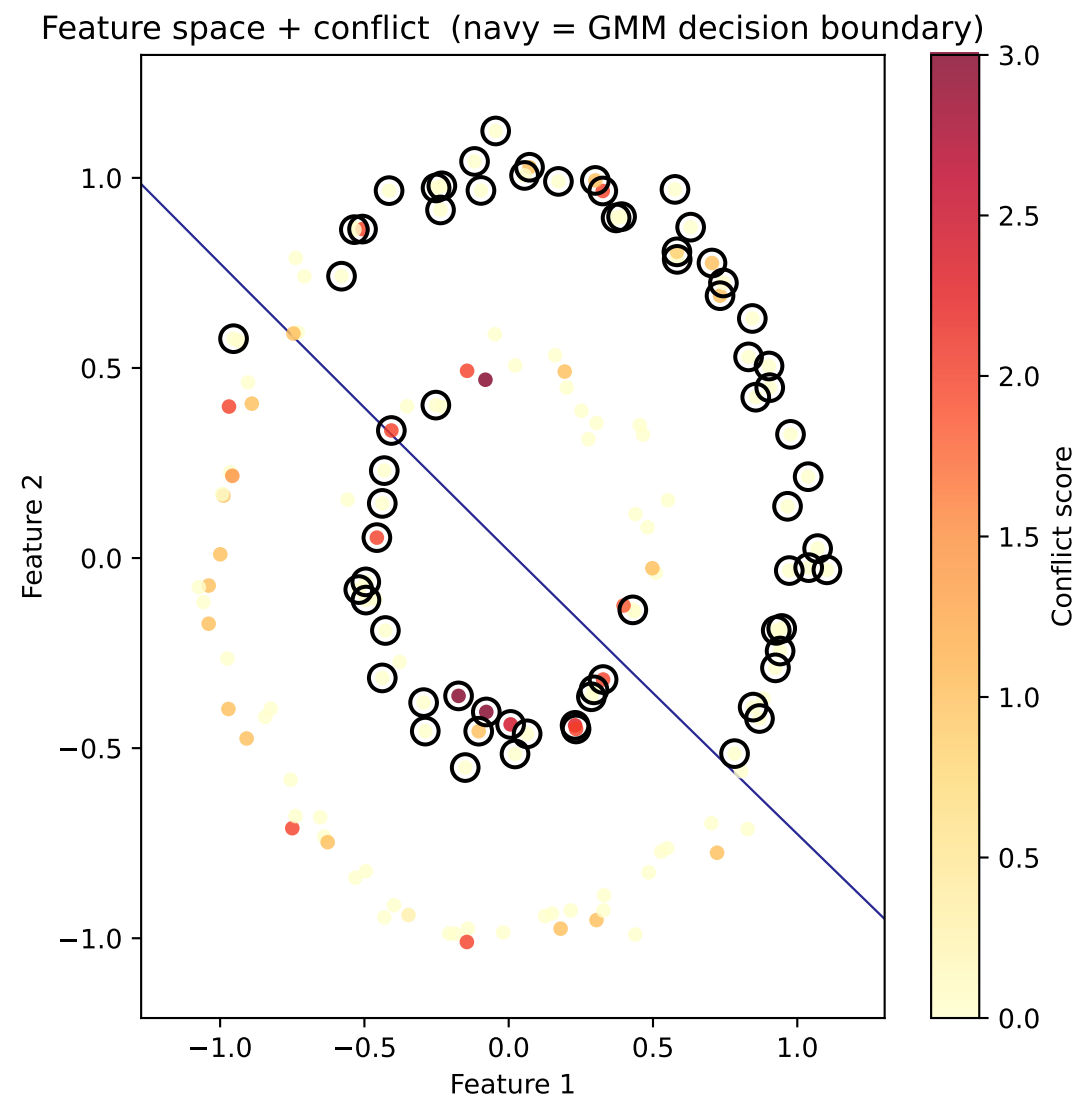
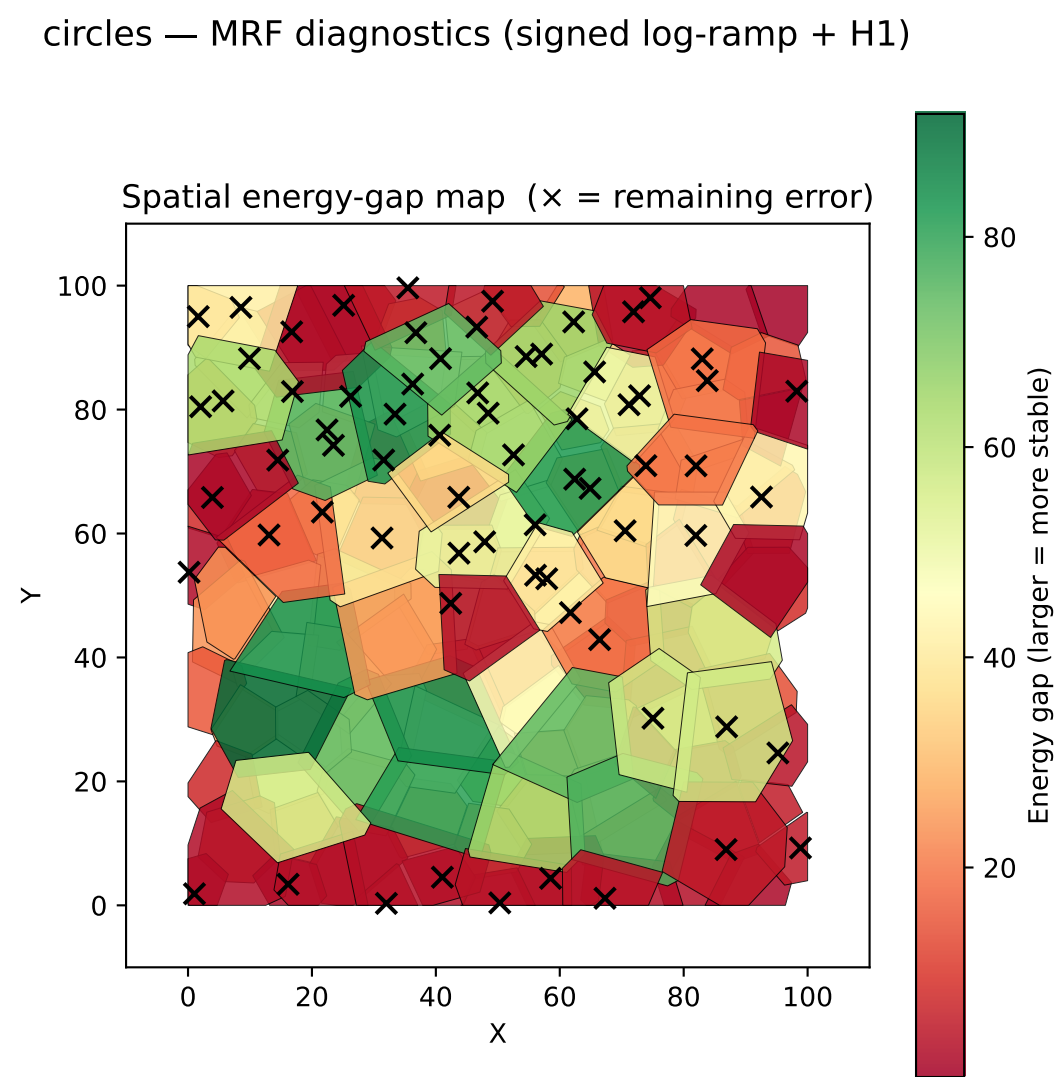
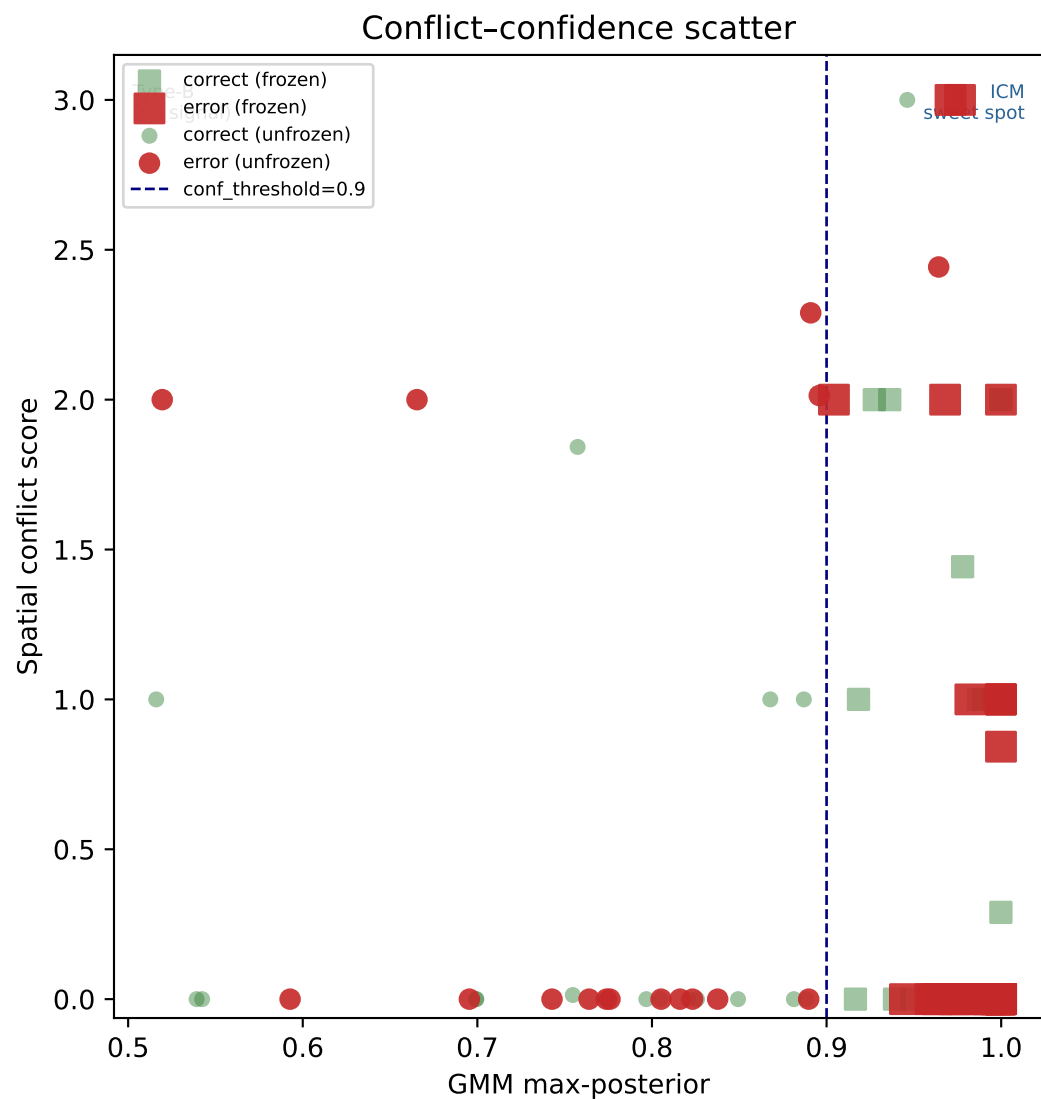


Cluster 1
68 cells (44 err)



Feature space





Run summary

MRF Run Summary	
ARI:	-0.0022
errors:	69 / 148
violations:	60 → 31
n_changed:	14
n_iters:	2
n_frozen:	111
Parameters:	
lam:	20.0
tau_low:	0.3
tau_high:	0.4
radius:	15.038613891552956
conf_thr:	0.9